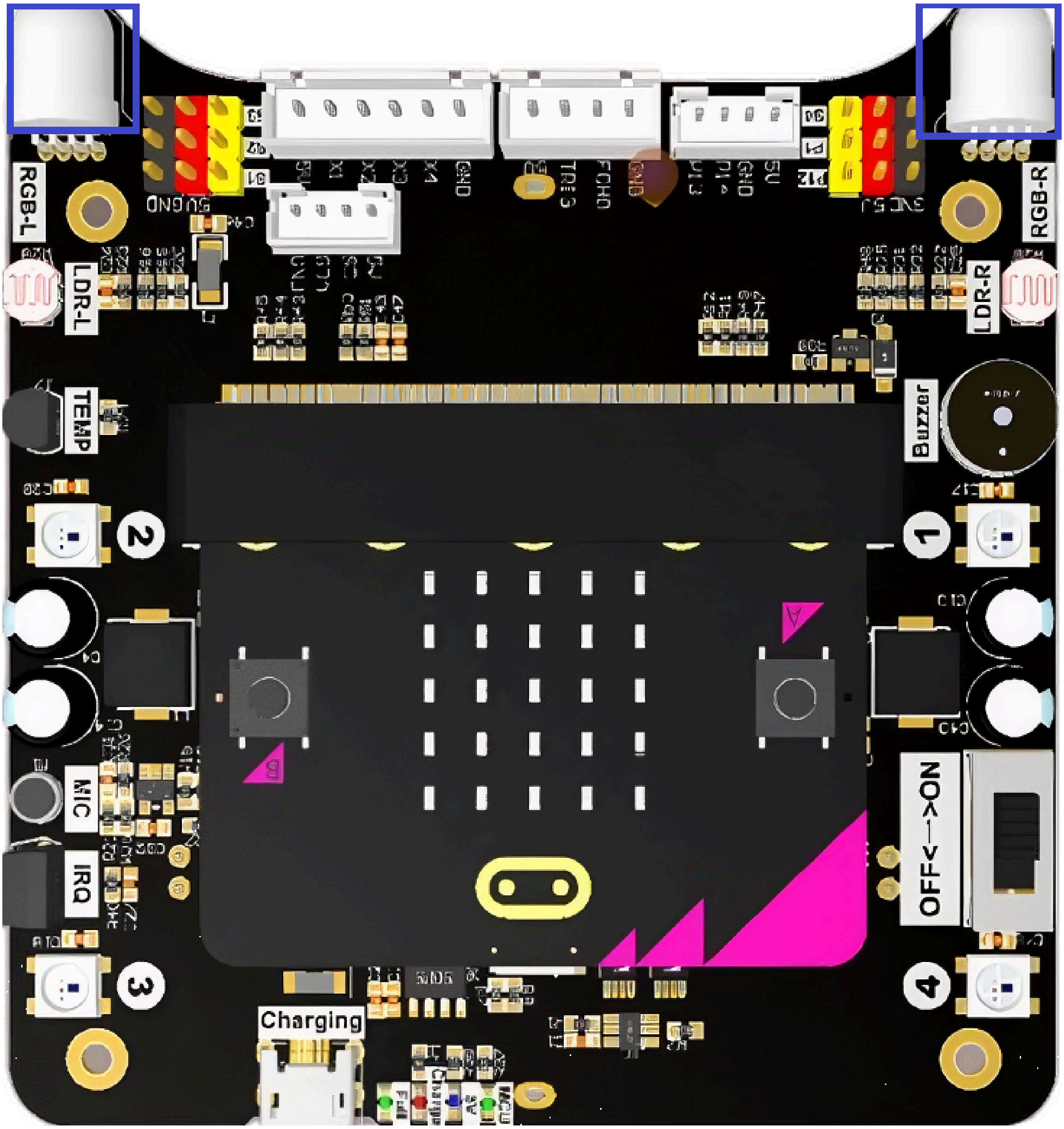
RGB Spotlight Alternating

1. Experiment Objective

The Lastbit car is equipped with two RGB spotlights on the front panel, RGB_L and RGB_R. In this section, you will learn how to light them up alternately to create a flowing light effect.

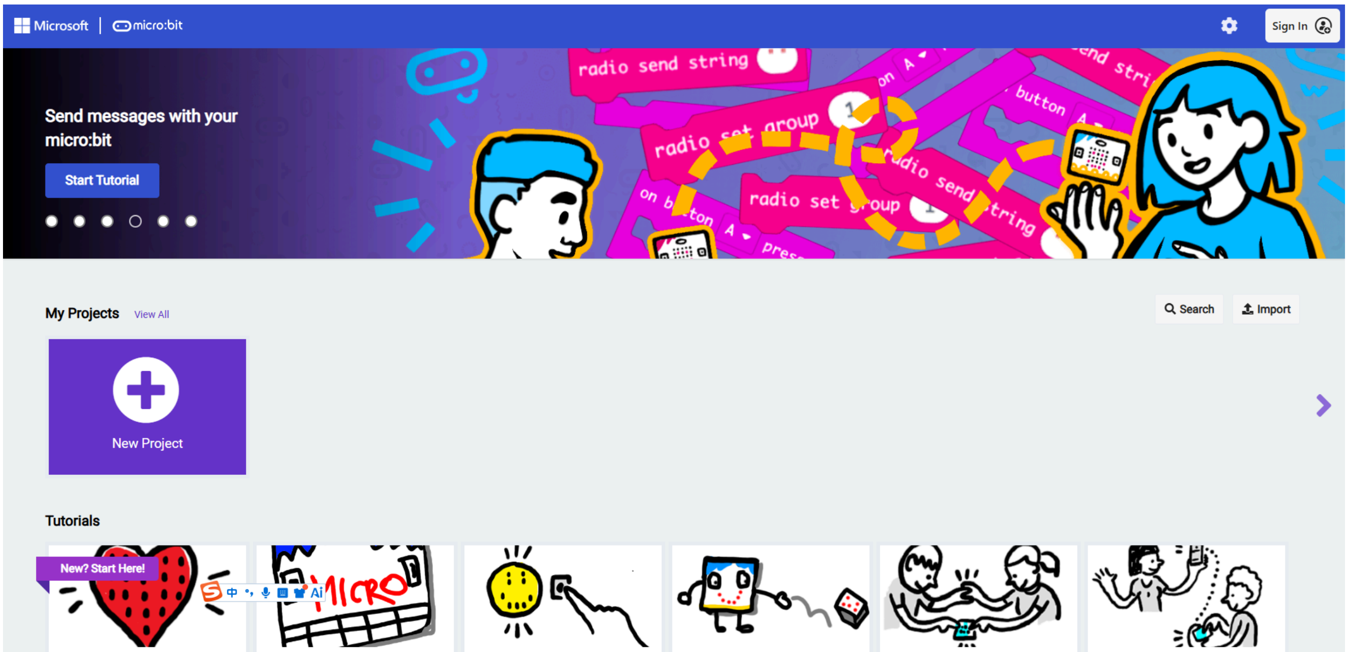


2. Experiment Steps

1. Before starting the experiment, connect the Micro:bit motherboard to your computer using a Micro USB data cable. At this time, the computer will display the MICROBIT drive letter.

2. Open MakeCode via the link below.

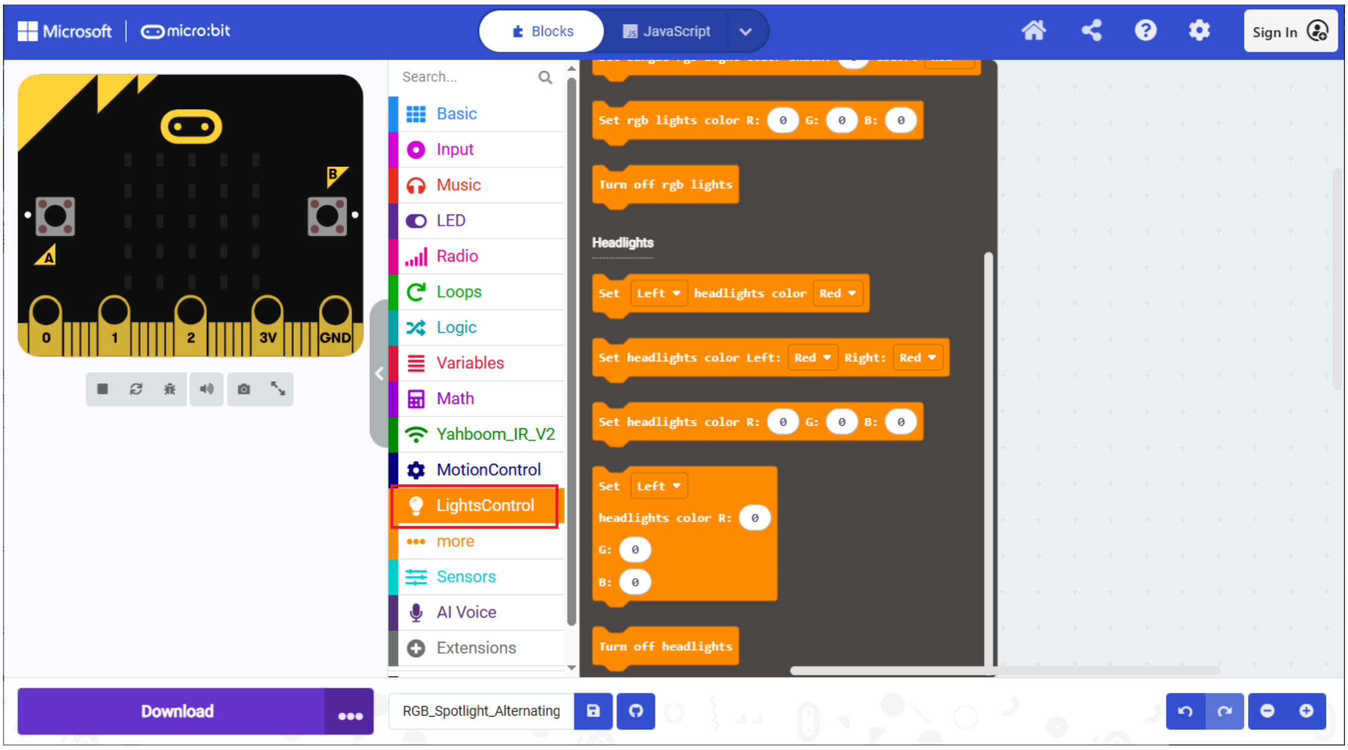
   [Microsoft MakeCode for micro:bit]()

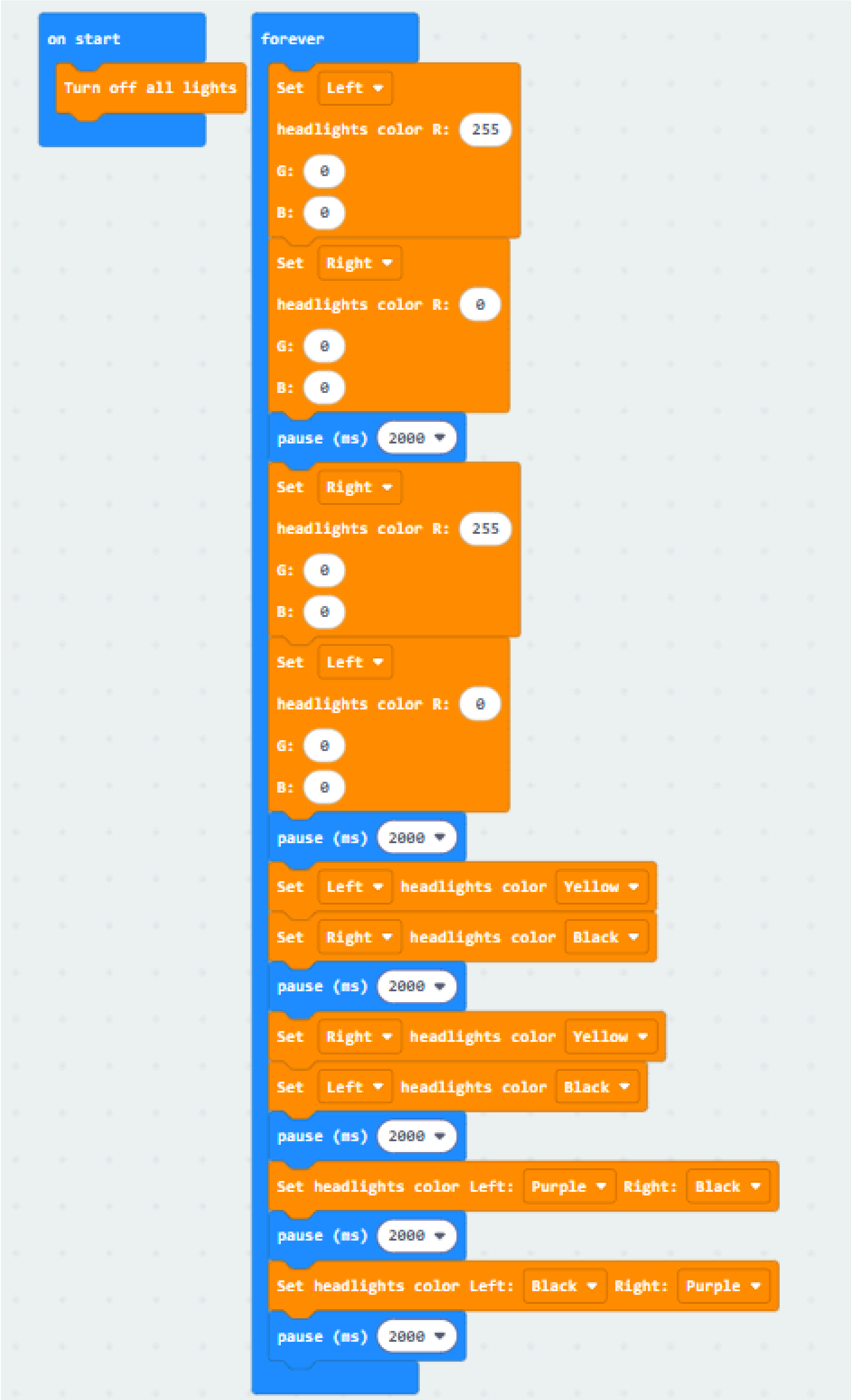


3. Drag `2.RGB_Spotlight_Alternating.hex` to the new project page, and MakeCode will automatically import and open the project.

3. Experiment Source Code

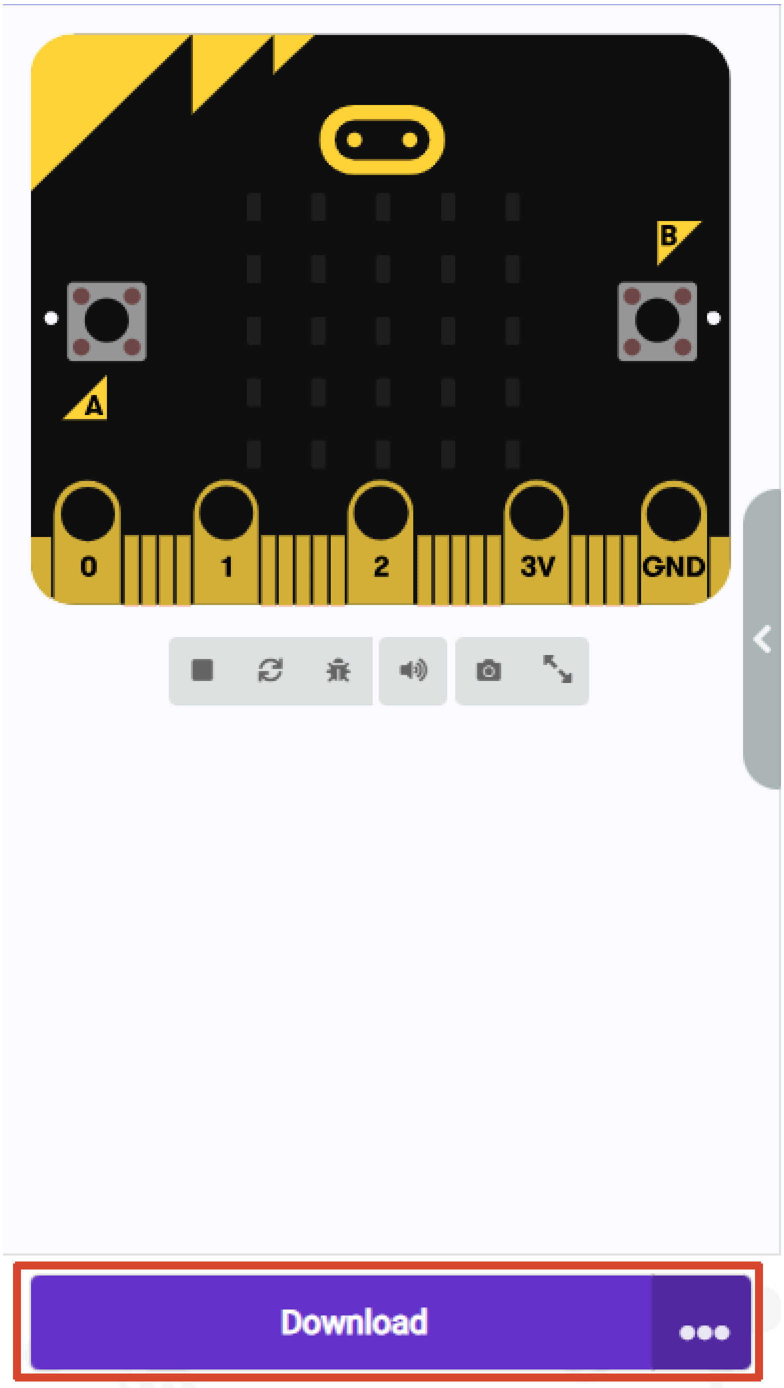
The building blocks for this section are in the `Headlights` option under the `LightsControl` category.



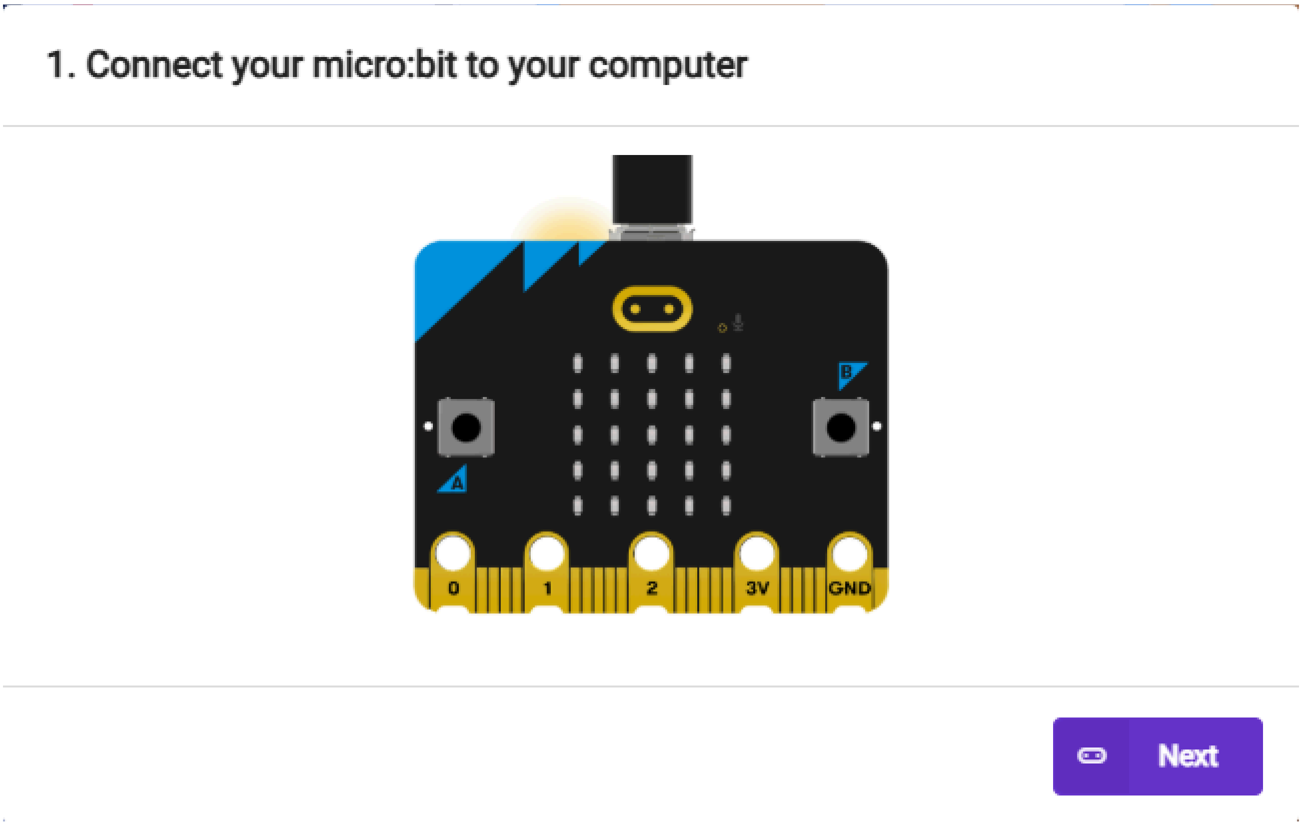


4. Program Download

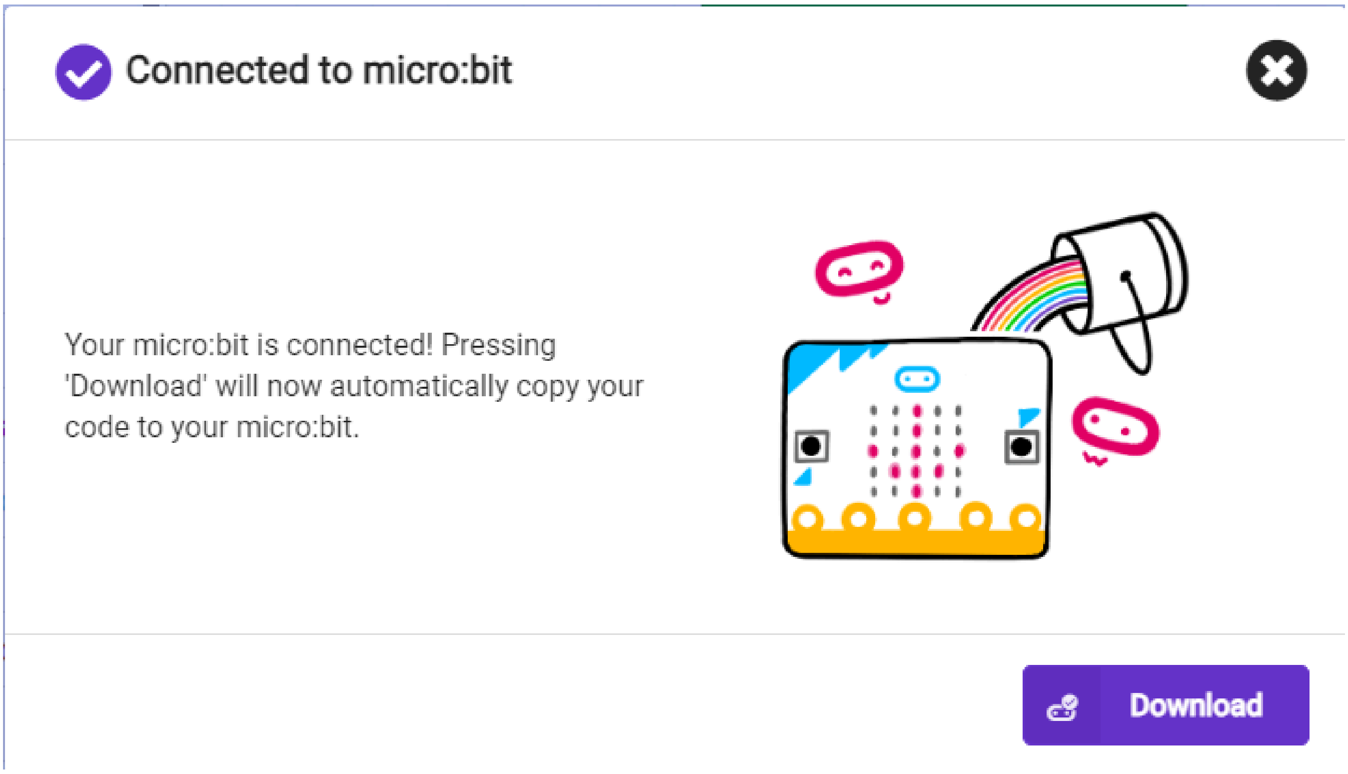
1. Click the download button at the bottom left to download the program to the Micro:bit.



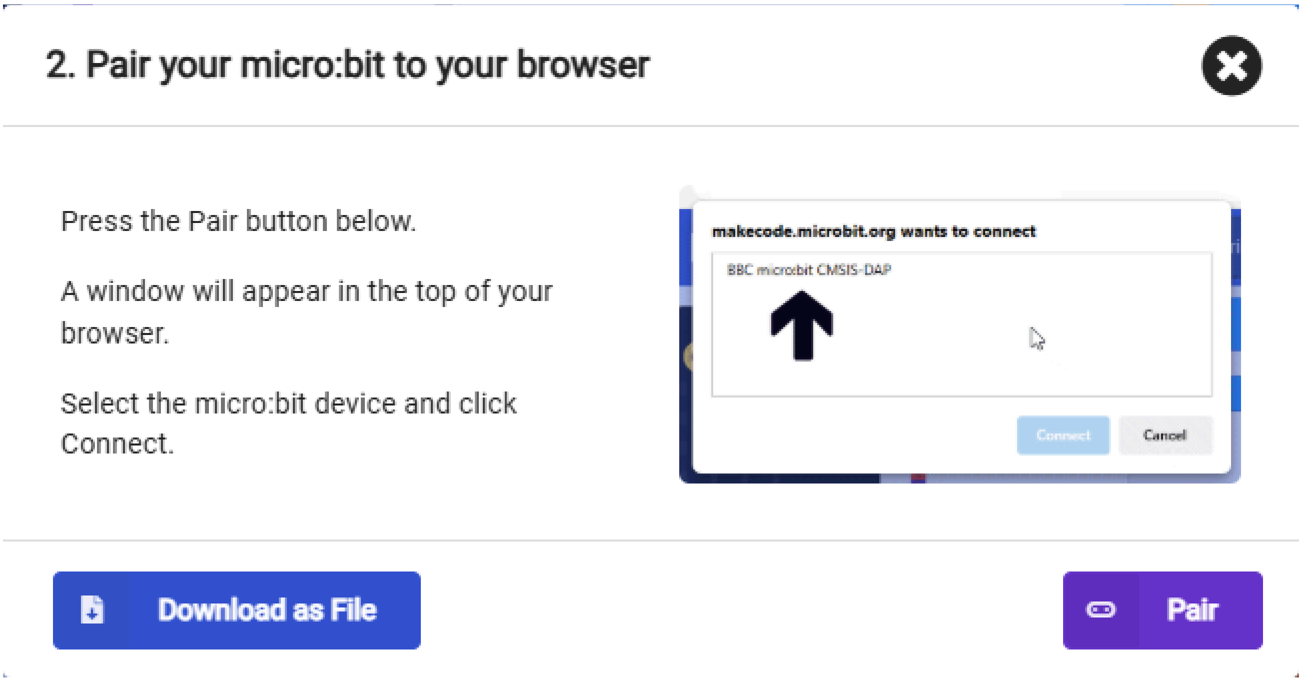
2. When the Micro USB data cable is not connected to the Microbit, the system will first prompt you to connect the device. If not connected, please first connect the data cable to the Microbit. If already connected, click 【Next】 here.



3. If the Microbit is detected to be properly connected, the following interface will appear. At this time, you can click 【Download】.

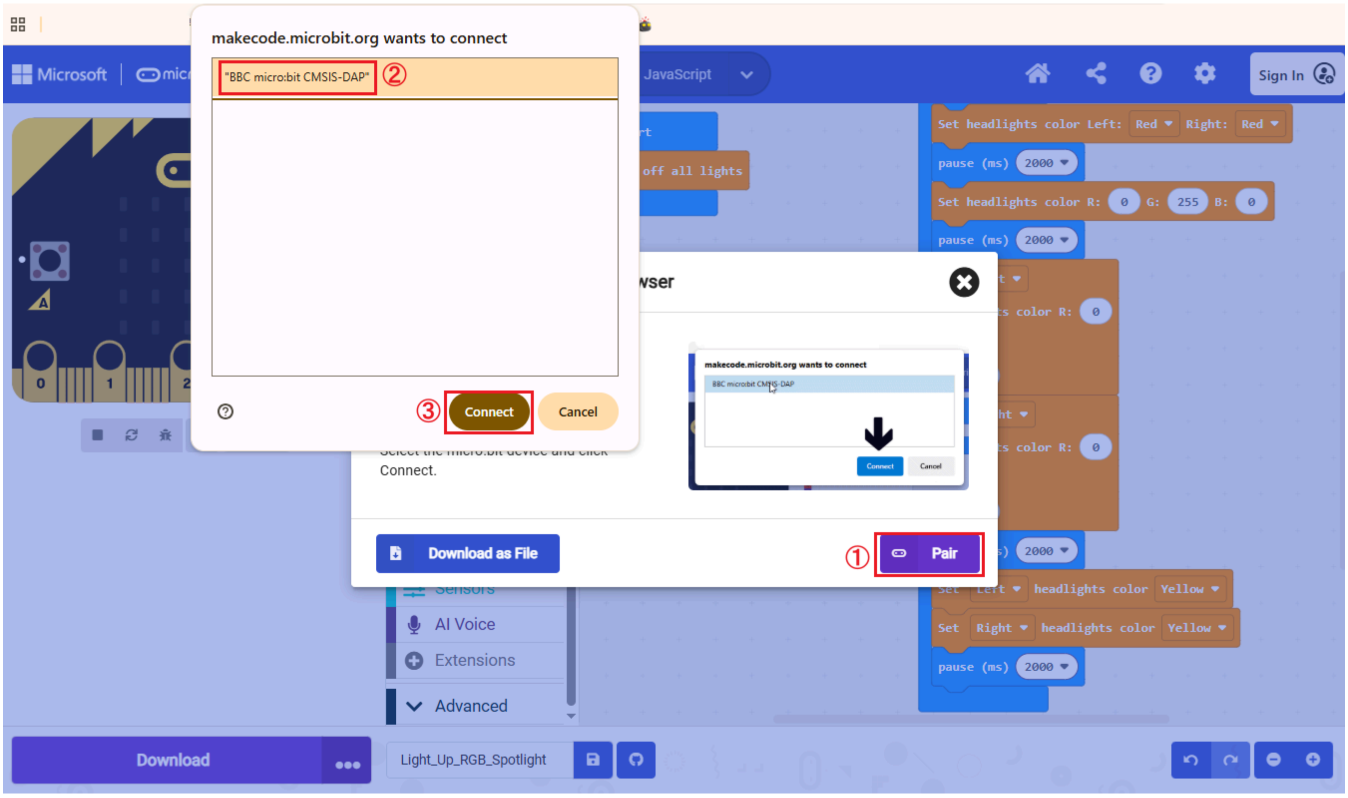


If the Microbit data cable is not properly connected or multiple Microbits are connected, the system will require you to manually select the device. Here you can choose two methods: 【Download as File】 and 【Pair】.

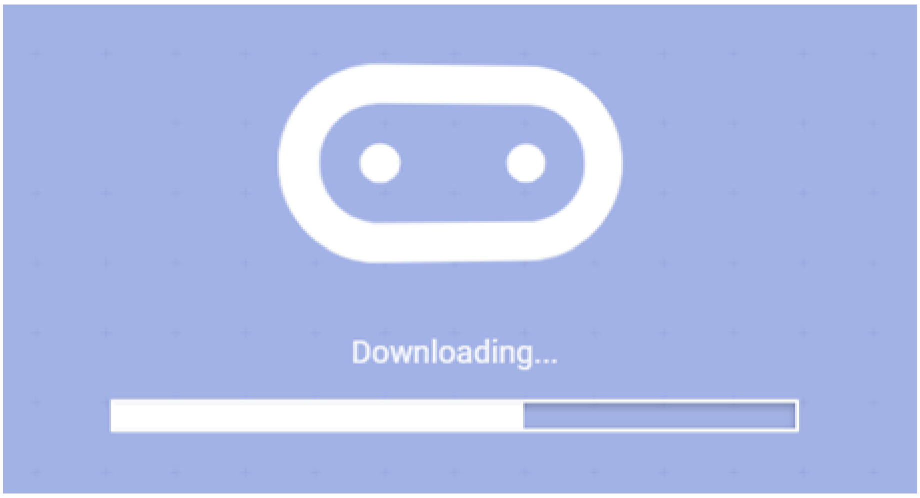


When selecting 【Download as File】, the system will download the currently opened program as `microbit-project-name.hex`. You can copy or drag this program directly to the corresponding drive letter of MICROBIT to complete the download.

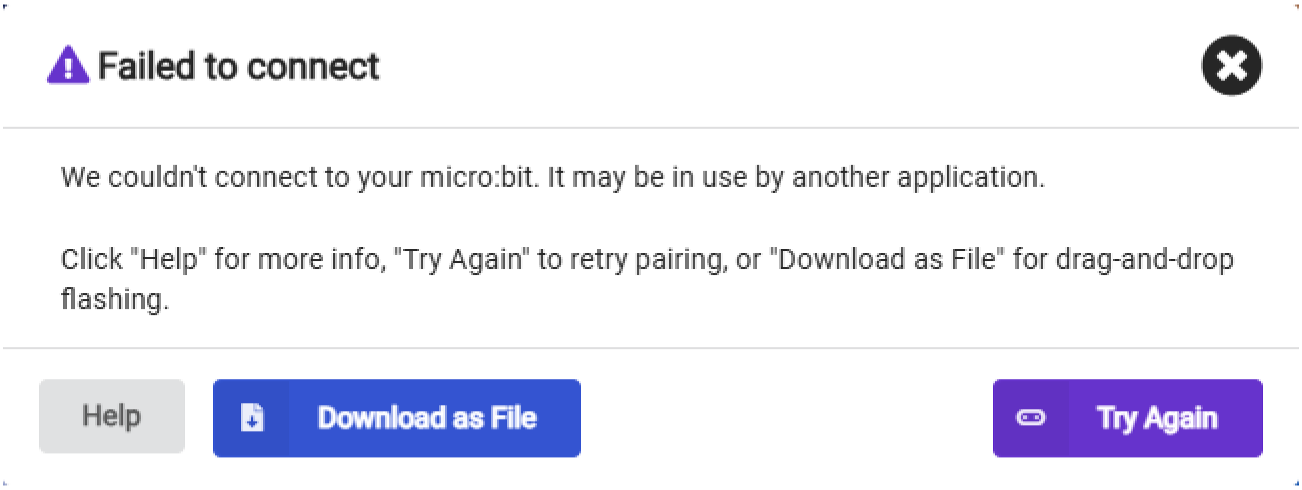
When selecting 【Pair】, you will need to manually select the device. Please select `BBC mico:bit CMSIS:DAP`.



4. When the following interface appears, the program is being downloaded. Wait for the progress bar to complete to download the program to the Microbit.



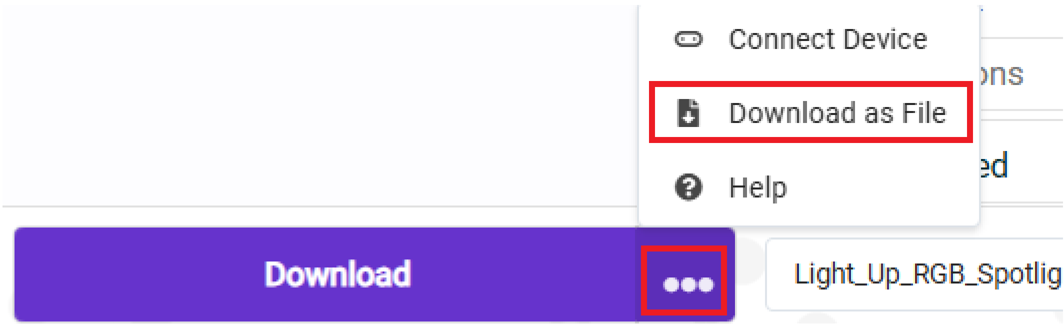
If multiple attempts always fail, directly select 【Download as File】 to save the current program as a file. Then copy or drag the program directly to the corresponding drive letter of MICROBIT to complete the download.



If you want to save the file directly, we provide two methods

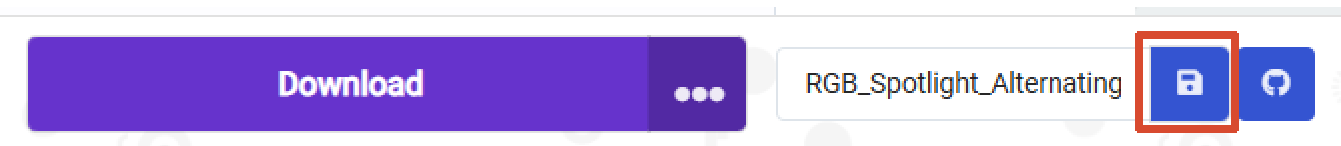
Method ①:

Click 【...】 next to download, then select 【Download as File】



Method ②:

Click 【Save】 after the project name edit field.



5. Experiment Results

The car spotlights will display in sequence: left red right off, right red left off, left yellow right off, right yellow left off, left purple right off, right purple left off. Each effect switches every 2 seconds and continues to cycle.

The colors used in this tutorial correspond to the following RGB values:

- Red: (255, 0, 0)
- Orange: (255, 128, 0)
- Yellow: (255, 255, 0)
- Green: (0, 255, 0)
- Blue: (0, 0, 255)
- Cyan: (0, 255, 255)
- Purple: (255, 0, 255)
- White: (255, 255, 255)
- Black (off): (0, 0, 0)